

Erythroblastosis fetalis

It is a hemolytic disease of fetus and newborn. Let's learn about the pathophysiology of this.

Level 1: meet the Red Blood Cell, tiny truck who delivered oxygen to whole body. For doing that they use a protein called hemoglobin. If lots of red cells are destroyed then our body can become short of oxygen carrying cells. Tiny cells, big jobs.

Level 2: blood groups.

A tiny tag on a red blood cell can make a big difference.

A tag

B tag

A+ B tag

No A+B tag /O tag

Another tag is Rh factor. If it presents then positive if not then negative. (luna: Rh factor is rhesus factor an inherited protein found on the surface of red blood cells that determines if your blood type is positive or negative.)

****picture match

Level 3: if mother is rh negative but fetus is positive then during pregnancy or delivery time little amount of blood introduce to mother's blood stream. The mother then makes anti D antibody. This antibody can cross the placenta in later rh positive pregnancy.

Level 4: mom makes antibody against the baby's blood group cause its foreign to herself. It called anti D. it crosses placenta and destroys baby's red blood cells.

Hemolysis means breakdown of red blood cells(RBC) which frees bilirubin(yellow pigment within hemoglobin) in blood stream and also develop anemia. Excess bilirubin in blood deposit under skin ,white part of our eyes is causing jaundice in neonate called neonatal jaundice.(luna: when bilirubin exceed 5-6mg/dl)

*** fill in the blanks:

1. red blood cells are commonly called ____ (rbc)
2. fetus and mother are connected through ____ (placenta)
3. a yellow pigment that can rise after breakdown of blood is ____ (bilirubin)
4. the blood tag involved in the classic case is ____ (Rh)

**** do you know the symptoms of jaundice and anemia? Let's guess.

- a. pale skin
- b. hyper action
- c. poor appetite
- d. tired and sick
- e. yellow discoloration of skin, sclera (luna), palm, tongue
- f. continuous cry
- g. rapid pulse (luna)
- h. playful
- i. Repeated breathing

level5: now we discuss about sign and symptoms of this.

1. anemia (luna: pallor (pale skin), tachycardia (fast heart rate), and tachypnea (rapid breathing))
2. jaundice: b (luna: yellow discoloration starts on the face, head, and sclera (whites of the eyes), then gradually spreads downward to the chest, abdomen, arms, legs, and soles of the feet), dark urine, clay colored stool.)
3. swelling (severe scenario)
4. kernicterus risk (luna: kernicterus is a rare severe form of brain damage caused by hyperbilirubinemia in newborn blood and brain tissue, when bilirubin exceeds the level of 25mg/dl)

Level 6** quick quiz: choose A,B OR C

1. What carries oxygen?

A. rbc

B. platelet

C. bone

2. which antibody can cross the placenta?

a. insulin

b. anti D

c. saliva

3. jaundice makes skin look-

a. blue

b. pink

c. yellow

level 7: prevention: preventive measures are the superhero move.

stop sensitization and protects future Rh+positive babies by Anti-D immunoglobulin injection to mother after proper guidance of her physician.

Timing of administration is always decided by clinicians.

**** figure****

Level 8: how is a baby helped? Exclusive breastfeeding if possible

Phototherapy(luna:special light helps lower bilirubin)

Blood transfusion(severe anemia)

Exchange transfusion and close monitoring after regular interval.

Level 9:

Can you want a small review:

The whole story in 6 steps.

Make a flow chart:

1. Rh negative mother may meet a rh positive blood fetus. 2. Immune system can make Anti-D. 3. Antibody can cross the placenta. 4. develop anemia-bilirubin-jaundice and many more endangering symptoms. 6. Prevention, early detect, protect babies.

Level 10.

**** q/a shuffle-match

1. rh positive mother and rh negative baby can leads to what immune problem?

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2. what happens to rbc in hemolysis?

They broke down

3. what yellow pigment can build up?

Bilirubin

4. name one prevention tool?

Anti D immunoglobulin

5. name the treatment of high bilirubin?

Phototherapy

Level 11:

Word matching game:

Hemolysis- Rbc breakdown

Jaundice -Yellowing

Placenta- Organ connecting mother and baby

Anemia- Low hemoglobin

Now we know about erythroblastosis fetalis and how we can defend it. Happy learning.